Sruthi Bijili

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.41
Sri Chaitanya Junior college	Class XII (SSC)	2024	98.00%
Jawahar navodaya vidyalaya	Class X (CBSE)	2022	91.20%

RELEVANT COURSES

Core Courses: Analog Electronics,Semiconductor Devices and Fundamentals,Digital Systems,Control Systems,Electrical Machines

Labs: Electric Devices and Circuits Lab,Digital Systems Lab,Electric Circuits Lab

Math Courses: Probability,Random Processes,Calculus,Intro to Statistics

Computer Science: Introduction to programming

SKILLS

Programming: Verilog, Python, C ,MATLAB, Latex

Tools:LTspice,Altium,Github,Linux,GTKWave

Concepts: Flip-Flops, Digital Electronics,Analog Electronics, Counters, FIR Filters

Platforms: HDLBits

PROJECTS

Microphone Amplifier with Class-AB Driver:

- Designed a low-noise microphone preamplifier using operational amplifiers.
- Implemented a Class-AB push-pull output stage to reduce crossover distortion and verified circuit performance using oscilloscope measurements.

Digital Clock System in Verilog:

- Designed a digital clock supporting time display and countdown timer.
- Implemented 24-hour format and verified functionality through simulation.

MOD-100 Counter using JK Flip-Flops:

- Designed and implemented a MOD-100 synchronous counter using 7476 JK flip-flop ICs by cascading two MOD-10 counters to generate a count sequence from 00 to 99.

Designing Various Circuits in LTSpice | Course Assignments:

- Designed and simulated various analog circuits in LTSpice as part of the Analog Electronics course.
- Implemented and analyzed circuits such as operational amplifiers, integrators, and differentiators.

Traffic light controller using verilog:

- Designed a finite state machine to control the traffic light sequence for multiple directions using Verilog HDL.
- Simulated and verified the timing sequence and state transitions to ensure correct traffic signal operation using GTK waves.

Arduino Scientific Calculator :

- Designed and developed a scientific calculator using Arduino UNO, a 16×2 LCD, and push-button inputs.

- Implemented arithmetic and trigonometric functions, and verified correct operation through hardware testing.

LCR meter PCB design:

- Designed the schematic and PCB layout for an LCR meter using Altium Designer.
- Performed component placement, PCB routing, and electrical design rule verification to prepare the layout for fabrication.